

RAFT-mediated graft polymerization of glycidyl methacrylate in emulsion from polyethylene/polypropylene initiated with γ -radiation

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ABSTRACT: The successful reversible addition-fragmentation (RAFT)-mediated graft polymerization of glycidyl methacrylate (GMA) in emulsion phase from polyethylene/polypropylene nonwoven fabric using 4-cyano-4-[(phenylcarbonothioyl)thio]pentanoic acid under γ -irradiation at ambient condition is reported. While conventional graft polymerization in emulsion phase yielded grafted materials with low degree of grafting (Dg) values [$<7.5\%$ at 10% (wt/wt) GMA], addition of RAFT agent to the graft polymerization system allowed the synthesis of polyethylene/polypropylene-g-poly(GMA) with more tunable Dg ($8\% \leq \text{Dg} \leq 94\%$) by controlling the grafting parameters. Relatively good control (polydispersity index ~ 1.2 for selected grafting conditions) during polymerization was attained at 100:1 monomer-to-RAFT agent molar ratio. The number average molecular weight of free poly(glycidyl methacrylate) (PGMA) increased as a function of monomer conversion. NMR analyses of the free PGMA homopolymers indicate the presence of dithiobenzoate group from 4-cyano-4-[(phenylcarbonothioyl)thio]pentanoic acid on the polymer chain. The reactive pendant oxirane group of the grafted GMA can be modified for various environmental and industrial applications. © 2017 Wiley Periodicals, Inc. *J. Appl. Polym. Sci.* 2017, 134, 45270.

KEYWORDS: emulsion polymerization; functionalization of polymers; grafting; irradiation

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INTRODUCTION

Surface modifications of polymers are performed to impart chemical groups with unique and advantageous properties that are essential for medical, biotechnology, catalysis, adsorption, and separation applications. Among the different techniques that have been employed for modifying polymer surfaces, graft polymerization of a functional monomer offers an effective approach for incorporating tunable surface properties, while retaining the desirable properties of the trunk material. Grafting can be initiated by chemical treatment, high-energy radiation, plasma-induced techniques, etc.¹ Radiation-induced grafting in particular has the advantages such as low cost, simplicity, control over process, and the possibility of grafting monomers that are difficult to polymerize by conventional methods without residues of initiators or catalysts.²

Electron beams and γ -radiation are examples of high-energy radiation that are used by researchers worldwide to initiate graft polymerization from solid polymer surfaces, which generally

proceeds via free radical polymerization process. However, irreversible termination of propagating polymer chains and chain transfer are features inherent to the mechanism of free radical polymerization. Termination of propagating radicals and chain transfer lead to a loss of control over chain architecture and chain length as well as broadening of the molar mass distribution.³ These limitations have led to the research field of controlled free-radical polymerization (CRP), and among the CRP techniques, chain transfer by reversible addition-fragmentation (RAFT) is one of the most versatile ones for providing living characteristics to free radical polymerization.^{3–5}

RAFT-mediated graft polymerization at ambient temperature by means of γ -radiation has been successfully performed for a variety of monomers.⁶ In this technique, organic compounds such as dithioesters and trithiocarbonates are used as chain transfer agents (CTAs). In the course of γ -initiated RAFT-mediated grafting, propagation of both grafted polymer (covalently linked to the surface) and free polymer (in the solution) are controlled by the same RAFT agent at the same time. Therefore, the

molecular weights and molecular weight distributions of grafted and free polymers should be very similar when the grafting occurs on the surface of the substrate.⁴

Conventional free radical polymerization is commonly employed in the commercial production of a wide range of synthetic materials. Up to 50% of these free radical polymerization processes are performed in emulsions. To gain acceptance in industry, RAFT-mediated polymerization and other CRP techniques must be successful in emulsion, particularly in water-based emulsion system.³ Besides being an environment friendly solvent, the use of water-based emulsions offers additional advantages such as easy processing of the product material. Though there are a number of publications dealing with synthesis of graft copolymers from synthetic and natural fibers through conventional radiation-induced grafting in emulsion phase,^{7–10} to the best of our knowledge there have been no report on γ -initiated RAFT-mediated graft polymerization of monomers in emulsion from synthetic polymers.

In this current work, we synthesized polyethylene/polypropylene-*g*-poly(glycidyl methacrylate) (PE/PP-*g*-PGMA) via the facile combination of RAFT-mediated grafting and γ -irradiation of PE/PP in glycidyl methacrylate (GMA) emulsion. The reactive oxirane group of GMA can be reacted with various reagents to impart new functionalities. This provides versatility to the grafted polymer that can be useful for different applications.

EXPERIMENTAL

Materials

The RAFT agent used in this study, 4-cyano-4-[(phenylcarbonothioyl)thio] pentanoic acid (CPPA, >97%), was purchased from STREM Chemicals, Massachusetts, USA. GMA (>95.0%, TCI) was deionized by percolation through a column of activated basic alumina (Ecochrom) and then stored at 4 °C. The surfactant, polyoxyethylene sorbitan monolaurate (Tween 20), tetrahydrofuran (THF, HPLC grade, >99.8%), and methanol (HPLC grade, >99.8%) were supplied by Kanto Chemical Co., Japan and were used as received. Nonwoven fabrics with fibers made of PP coated with PE (PE/PP) were provided by Kurashiki Sen-I Kako Co. Okayama, Japan and were used as polymeric substrates for the grafting experiments.

Irradiation

The ⁶⁰Co irradiation facility in Takasaki Advanced Radiation Research Institute, Japan was used for the irradiation of samples at room temperature and with a dose rate of 1 kGy h⁻¹. Various absorbed doses, i.e., 0.5, 1, 3, and 6 kGy, were applied throughout the study.

Graft Polymerization

In a typical RAFT-mediated grafting, a piece of PE/PP with approximately 1 cm × 3 cm dimensions and a weight of ~0.02 g was immersed into each grafting solution (~8 g) prepared by homogenizing GMA, CPPA, and Tween 20 in water at 6000 rpm. The polymerization mixture in glass vial capped with self-sealing septum was connected to N₂ bubbling at room temperature for 10 min, and then placed in γ -irradiator chamber. After the graft polymerization was carried out at room temperature for the prescribed time, the polymerization was stopped by

removing it from the ⁶⁰Co irradiation room. PE/PP-*g*-PGMA was washed with THF overnight to remove free homopolymers and ungrafted GMA monomer. The PE/PP-*g*-PGMA was then dried to constant weight under vacuum at 35 °C. The degree of grafting (Dg, wt %) was calculated using the following equation:

$$Dg = \frac{W_f - W_o}{W_o} \times 100 \quad (1)$$

where W_o (g) is the weight of pristine PE/PP and W_f (g) is the dry weight of PE/PP-*g*-PGMA. Conversion (%) was defined as the fraction of the monomer (GMA) polymerized and was determined gravimetrically after vacuum filtration, through a 0.45- μ m PTFE membrane filter, of the polymerization mixture that was precipitated in fivefold volumes of methanol. The PGMA homopolymers on the PTFE membrane filter was dried initially in fume hood and then in a vacuum oven. The grafted PGMA on the PE/PP trunk material also contributed to the overall conversion and was included in the calculation. The experiments were repeated to check data precision and the maximum standard error ($n=9$) calculated was 2%. Gravimetric measurements were carried out using a Sartorius CPA 225D model balance with a sensitivity of ± 0.01 mg.

The theoretical number-average molecular weight, $M_{n,th}$, was calculated according to the following equation:

$$M_{n,th} = M_{CTA} + \frac{n_m^0}{n_{CTA}^0} \times \text{conversion} \quad (2)$$

where $M_{n,th}$ is the theoretical number average molecular weight of the polymer, n_m^0 , the initial number of moles of the monomer, i.e., GMA, present in the system, n_{CTA}^0 , the initial number of moles of CTA, i.e., RAFT agent, CPPA, present in the system and M_{CTA} , the molecular weight of the CTA. In order to study the features of γ -ray initiated RAFT-mediated grafting in emulsion state, conventional grafting studies were also performed where PE/PP substrates were grafted with PGMA identically with the exception that no RAFT agent was added to the emulsion.

Gel Permeation Chromatography (GPC) Analysis

The molecular weights and polydispersity index (PDI) of the free PGMA homopolymers were measured by GPC on a Hitachi GPC instrument equipped with a Shodex 9 μ m particle size guard column (50 × 7.5 mm) followed by two Shodex GPC columns (300 × 8.0 mm) in series, L-7100 solvent delivery system, L-7490 RI detector, and L-7400 UV detector. The guard and GPC columns were kept in a Shimadzu column oven at 40 °C during the analysis. The eluent was THF (1 mL min⁻¹ flow rate) and the system was calibrated with polystyrene standards (Polysciences, Inc., Pennsylvania, USA) with molecular weights of 150–10⁶ g mol⁻¹.

Attenuated Total Reflectance (ATR) Fourier Transform Infrared (FTIR) Spectroscopy

The infrared spectra of PGMA homopolymer, pristine PE/PP, and PE/PP-*g*-PGMA were examined by a Perkin Elmer Spectrum One FTIR spectrophotometer in ATR mode. The samples were scanned in the range 650–4000 cm⁻¹ with a resolution of 4 cm⁻¹. Peak area calculations were carried out using the Spectrum software.

Scanning Electron Microscopy (SEM) Imaging and Energy Dispersive X-ray (EDX) Mapping

SEM images and oxygen elemental maps were taken using a Hitachi SU 3500 (Hitachi, Japan) scanning electron microscope equipped with an EDX spectrometer (X-Max, Horiba). Pictures of the fibers' surfaces were acquired at low vacuum condition and acceleration voltage of 5 kV while the oxygen elemental maps were taken at an acceleration voltage of 8.5 kV.

Thermogravimetric Analysis

Thermal decomposition properties of the pristine PE/PP and PE/PP-g-PGMA were recorded using a TG/DTA 6200 Exstar 6000 (Seiko Instruments Inc., Japan). Analyses were conducted over the temperature range 25–500 °C with a programmed temperature increment of 10 °C min⁻¹ under dynamic nitrogen atmosphere.

Differential Scanning Calorimetry Analysis

The glass transition temperature, T_g , of the homopolymers and the high Dg samples was determined on a Thermo Plus DSC8230 (Rigaku, Japan). Each sample was packed in a standard aluminum pan that was hermetically sealed. The samples were studied under dynamic nitrogen atmosphere using a cyclic three-stage thermal program. The first stage was a heating at a constant rate (10 °C min⁻¹) from 35 to 95 °C. At the end of this stage, the samples were cooled at -2 °C min⁻¹ to 35 °C (second stage) and then heated again to 95 °C (third stage) at a constant rate (10 °C min⁻¹). The T_g was determined from the second heating curve using the Thermo Plus 2 software.

X-ray Photoelectron Spectroscopy (XPS)

X-ray photoelectron spectra were recorded on a PHI 5000 Versa Probe II Scanning XPS Microprobe with monochromatized Al K α X-ray source (1486.6 eV photons) at a pass energy of 29 eV for region scan spectra and 117 eV for survey scan spectra. The anode current was 3.0 mA. The pressure in the analysis chamber was maintained at 1×10^{-6} Pa or lower during each measurement. The samples were mounted on the standard sample holder by means of indium and carbon adhesive tapes. The core level signals were measured at the photoelectron takeoff angle of 45°. All binding energies were referenced to the C1s hydrocarbon peak at 284.8 eV. Surface elemental stoichiometries were determined from peak-area ratios, after correcting with experimentally determined sensitivity factors as set by the instrument manufacturer.

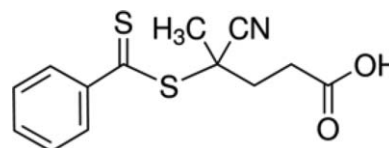
Proton Nuclear Magnetic Resonance Analysis

The proton nuclear magnetic resonance analysis (¹H-NMR) spectrum of the homopolymer was measured using a Bruker Avance III 300 Nuclear Magnetic Resonance Spectrometer. The sample was dissolved in deuterated chloroform (99% purity, 99.8% deuteration degree, Kanto Chemical Co. Inc., Japan) with tetramethylsilane as internal standard. Peak assignments and integration were performed using Bruker Top Spin 1.3 software.

RESULTS AND DISCUSSION

RAFT-Mediated Synthesis of PE/PP-g-PGMA

Pan and coworkers first reported the controlled radical polymerization under constant source of γ -radiation in the presence of



Scheme 1. CPPA RAFT agent.

thiocarbonylthio compounds in 2001.^{11–13} Since these initial reports, there has been a considerable effort given to radiation-induced RAFT-mediated homo- and copolymerizations of a large number of monomers.^{2–6,14,15} In their paper, Barner *et al.*, proposed a reaction mechanism for RAFT graft polymerization in solution under a constant source of γ -radiation, wherein the basic steps of RAFT mechanism are extended by reaction steps that consider grafting onto polymeric surfaces.¹⁴ In the simultaneous grafting technique employed during the grafting studies, initiating radicals were generated via application of γ -radiation. Radicals can be formed at PE/PP surfaces and in the monomer emulsion. The radicals formed on the PE/PP surface served as points of initiation and growth for the graft chains. At the same time, γ -radiation also initiated homopolymerization of GMA in the emulsion. Both propagating chains, i.e., grafted and free in mixture, then add to the thiocarbonyl group of the CPPA, creating the RAFT radical intermediates. Upon addition of the propagating chains to CPPA molecules, the leaving group radical was released from CPPA and subsequently re-initiated the polymerization steps in the mixture. These reaction steps comprised the pre-equilibrium part of RAFT mechanism. The core steps of the RAFT process is the main equilibrium wherein the free macro radicals that are either covalently bonded to the PE/PP surface or are free in the reaction mixture are adding to the polymeric RAFT agent to form macroRAFT intermediate radicals which in turn may fragment by releasing the starting materials or the formerly attached polymeric entity. Because RAFT-mediated graft polymerization is not a true living polymerization, termination of propagating chains is still a possibility and this may occur via bimolecular termination reactions between free or surface attached macro radicals.

In order to study the RAFT-mediated graft polymerization of GMA onto PE/PP, the CPPA RAFT agent was employed in this study (Scheme 1). CPPA has a carboxylic group which allows good solubility in aqueous solutions and it had been shown to control the polymerization of methacrylate monomers, such as 2-(dimethylamino)ethyl methacrylate and oligo(ethylene glycol) methacrylate, in aqueous media with addition of small amount of organic solvent.¹⁶ Reaction conditions, i.e., monomer concentration, RAFT agent concentration, monomer to RAFT agent ratio and absorbed dose, also affect the RAFT process and determine the observed Dg values. The effects of these grafting parameters on the RAFT mediated graft polymerization of GMA onto PE/PP were investigated and the results are summarized in Table I.

In conventional graft polymerization in emulsion state under γ -radiation, the monomer, polymer substrate, and water are all exposed to radiation and thereby produce radical species.

Table I. RAFT-Mediated Graft Polymerization of GMA Aqueous Emulsion (10:1 GMA to Tween 20 Weight Ratio) Onto PE/PP at Room Temperature Under γ -Irradiation at Dose Rate of 1 kGy h⁻¹

Expt.	C _{GMA,o} (wt %)	C _{CPPA,o} (wt %)	n _m ⁰ /n _{CTA} ⁰	Dose (kGy)	Dg ^a (%)	Conv. ^b (%)	M _{n,Th} ^c (g mol ⁻¹)	M _{n,GPC} ^d (g mol ⁻¹)	PDI ^d	M _{n,NMR} ^e (g mol ⁻¹)
1	3.0%	0.06%	100	0.5	8.2	38	5720	4600	1.43	7800
2	3.0%	0.06%	100	1	9.5	85	12,386	6000	1.62	16,600
3	3.0%	0.06%	100	3	10.4	90	13,058	5900	1.86	N/A ^f
4	3.0%	0.015%	400	0.5	6.6	80	46,150	34,500	4.66	N/A ^f
5	3.0%	0.015%	400	1	6.6	86	49,363	37,500	7.28	N/A ^f
6	5.0%	0.10%	100	0.5	19.2	11	1886	2700	1.16	2000
7	5.0%	0.10%	100	1	23.6	90	13,140	5900	1.35	15,900
8	5.0%	0.10%	100	3	26.5	91	13,376	5800	1.47	N/A ^f
9	5.0%	0.025%	400	0.5	14.6	77	43,502	30,000	3.83	43,100
10	5.0%	0.025%	400	1	16.5	89	50,456	31,300	4.35	63,900
11	10.0%	0.20%	100	0.5	26.9	8	1413	2500	1.16	2400
12	10.0%	0.20%	100	1	82.3	53	7844	4900	1.16	8600
13	10.0%	0.20%	100	3	92.7	94	13,613	6700	1.37	13,300
14	10.0%	0.20%	100	6	93.7	93	13,531	6500	1.32	15,900
15	10.0%	0.05%	400	0.5	36.0	24	13,643	7800	1.29	11,100
16	10.0%	0.05%	400	1	43.7	90	51,342	13,600	1.28	51,100
17	10.0%	0.05%	400	3	45.0	92	52,195	13,600	1.38	N/A ^f
18	3.0%	—	—	0.5	1.3	93	—	179,600	7.29	N/A ^f
19	3.0%	—	—	1	2.5	93	—	184,100	7.30	N/A ^f
20	10.0%	—	—	0.5	7.1	95	—	123,000	9.51	N/A ^f
21	10.0%	—	—	1	7.4	96	—	108,000	9.53	N/A ^f

^aDg calculated using eq. (1).^bConversion of monomer calculated gravimetrically.^cTheoretical number-average molecular weight was calculated from the monomer conversion using eq. (2).^dNumber average molecular weight, M_{n,GPC}, and PDI are determined from PGMA homopolymers formed in the polymerization mixture and were obtained from GPC analysis.^eNumber average molecular weight determined from NMR spectrum of the PGMA homopolymers formed in the polymerization mixture and calculated using eq. (3).^fNot available.

Irradiation of PE resulted to bond-breaking reactions that lead to production of free radicals, majority of which were allyl, alkyl, and polyenyl type.^{17,18} The relative amount of these carbon-centered free radicals depends on the temperature, energy, type of irradiated polymer and type of radiation, and post-irradiation storage time and environment. These radicals served as initiation points for graft polymerization of GMA monomer on the surface of PE/PP while the free polymers in the solution were initiated by the radicals produced upon interaction of GMA and water molecules with γ -radiation. Entries 18–19 and 20–21 give the results after irradiation of PE/PP in 3% and 10% GMA emulsion without the RAFT agent, i.e., conventional grafting, respectively. These experiments were carried out by γ -irradiation of the samples to 0.5 and 1.0 kGy at a dose rate of 1 kGy h⁻¹. The calculated conversions in these experiments were greater than 90%, indicating a very high rate of polymerization. The combination of the free radical process, provided by the γ -irradiation, and the emulsion technique was known to result in high conversion and high reaction rates of the polymerization process.³ Due to fast polymerization

kinetics, the effect of absorbed dose on Dg in these experiments was not obvious. The Dg increased with increasing monomer concentration, i.e., at 1 kGy absorbed dose, the calculated Dg's were 2.5% and 7.4% for 3% and 10% GMA, respectively. This was due to an increase in monomer concentration in close proximity to the substrate grafted, which gave higher chance to the monomer to react with the propagating graft chain.² Khan *et al.* achieved almost the same Dg values when they grafted 10% (vol/vol) methyl methacrylate, which was emulsified with nonionic surfactant octylphenoxy-polyethoxyethanol (Triton X-114), from Jute fibers using conventional simultaneous grafting technique under γ -radiation.¹⁹

The Dg's of PE/PP-g-PGMA after γ -irradiation of the polymer substrate in GMA emulsion in the presence of the RAFT agent are shown as entries 1–17 in Table I. Comparison of the entries with same monomer concentration and irradiated at the same absorbed dose, with and without RAFT agent (i.e., entries 2 and 5 with entry 19; entries 12 and 16 with entry 21), showed that addition of RAFT agent to the polymerization medium resulted in increased Dg. Without RAFT agent, i.e., conventional

grafting, the polymerization proceeds via free radical polymerization and irreversible termination of growing radicals is a feature inherent to this process. Addition of RAFT agent decreased the amount of propagating radicals, both on the PE/PP surface and in the bulk of polymerization medium, by reversible deactivation. This reduced the probability of irreversible termination, since the rate of termination is strongly dependent on the active radical concentration. The suppression of irreversible termination reactions allowed a “living” system where the graft chains propagated longer,^{2–4} resulting to higher Dg. Kodama *et al.* also observed a similar trend when they graft polymerized 2-hydroxyethyl methacrylate from PE/PP in the presence of cumyl dithiobenzoate RAFT agent under γ -radiation.²⁰ They attributed the higher Dg observed in the presence of RAFT agent to the longer irradiation time of samples with cumyl dithiobenzoate. In the present study, the irradiation time for both conventional and RAFT-mediated graft polymerization of GMA in emulsion phase were similar, therefore the observed trend can be attributed solely to the presence of RAFT agent and the reactions it induced in the graft polymerization system as discussed above.

The Dg of PE/PP-g-PGMA from the RAFT-mediated graft polymerization exhibited dependence on absorbed dose; unlike the results obtained from conventional grafting. The Dg rapidly increased after irradiation to 0.5 kGy, then leveled off and the Dg did not increase significantly (i.e., entries 11–14 and 15–17). Higher absorbed dose produced more free radicals on the substrate, which initiated more graft chains. As the polymerization proceeded, the monomer conversion increased while the concentration of monomer available for grafting decreased. The two effects counteracted one another, allowing the Dg to reach a stable value and then levels off.² In addition, the increase in Dg may also be attributed to increase in molecular weight of the graft chains. The RAFT process provided living character to free radical polymerization so the grafted PGMA chains were expected to continue propagating through the polymerization time. It is well known that a reduction in rate of polymerization was often observed in chain transfer polymerization, including RAFT-mediated polymerization, so lower monomer conversion was normally observed at the early portion of the polymerization process.²¹ In the case of conventional grafting, the conversion reached a very high value, >90%, at 0.5 kGy; hence, further production of free radicals by direct and indirect interaction of γ -radiation with the PE/PP substrate did not result to further increase in Dg because of monomer exhaustion.

Increasing the monomer concentration in the RAFT-mediated process increased the Dg of the grafted PE/PP (entries 1–3, 6–8, and 11–13). Again, this is simply because of closer proximity with the free radicals generated on the PE/PP substrate and propagating polymer chains. The Dg increased with decreasing monomer to RAFT agent ratio (i.e., entries 1–2 and 4–5, 6–7, and 9–10, 12–13, and 16–17). This is contrary to the results found by other researchers wherein the Dg increased with increasing monomer to RAFT agent ratio.^{20,22} They attributed the observed trend to higher monomer conversion and molecular weight values of graft chains obtained at the same grafting time when the RAFT agent concentration was lower. In the present study, the molecular weights of the graft chains also

increased with decreasing RAFT agent concentration, as shown in Table I. The observed trend in this study might be attributed to the indirect suppression of the termination reactions in solutions containing high amounts of the RAFT agent, i.e., low monomer to RAFT agent ratio. The indirect suppression of the termination reactions possibly lead to more propagating grafted chains, resulting in higher Dg values for solutions with lower monomer to RAFT agent ratios.^{2–4}

The techniques used in this study allowed the synthesis of PE/PP fibers functionalized with oxirane groups from the PGMA chains, grafted in a controlled manner by a one-step method that lead to robust and stable covalent bond between the surface and grafted PGMA at room temperature and without any pre-functionalization step. Table I shows that the Dg can be controlled from 8% to 90%, with graft chains of reasonable dispersity ranging from 1.2 to 1.5, by varying the monomer concentration, absorbed dose, amount of CPPA, and monomer to RAFT agent ratio. The optimum Dg value depends on the intended application of the final material. For preparation of biocompatible surface grafting of 2%–4% may be sufficient, for fuel cell and battery separator applications grafting of 20%–50% would be desired and for adsorption applications even grafting of 100% would be preferred.²³ Further chemical modifications of the PGMA graft chains impart new functional groups to the PE/PP grafted fibers for various applications.

Characterization of the Homopolymers Formed in the Polymerization Mixture During Grafting

In the graft polymerization experiments performed in this work, coagulum formation was rarely observed and the highly colored creamed oils observed on the top of the reaction mixtures, as reported by other authors,^{24,25} were not seen at any stage during the polymerization experiments. Application of γ -radiation generated radicals on the PE/PP surface and in the monomer emulsion. Subsequently, radicals from monomer and water and radicals formed on the surface initiated the propagating chains, which then added to the thiocarbonyl group of the RAFT agent, creating the RAFT radical intermediates. Because both the grafted chains and free PGMA were controlled by the same RAFT agent at the same time, the molecular weight and molecular weight distribution of the grafted and free PGMA should be very similar when the grafting occurred on the surface of the substrate.⁴ Therefore, analyses of the free PGMA in the emulsion not only give us information on the homopolymer part of the grafting medium but also allow us to estimate the molecular weight and dispersity of the grafted chains. The effects of absorbed dose, monomer concentration, CPPA concentration, and monomer-to-RAFT agent ratio on the molecular weight and molecular weight distribution of the free PGMA are shown in Table I. The controlled fashion for the RAFT-mediated polymerizations is shown in this table and corresponding figure, Figure 1, which depicts the molecular weight distribution of free PGMA at different absorbed doses and monomer-to-RAFT agent ratio. Symmetric peaks and narrow molecular weight distributions can be observed from peaks (a) and (b). These peaks were obtained from samples that have low monomer-to-RAFT agent ratio (i.e., high RAFT agent concentration) and irradiated at low absorbed dose. Increasing either

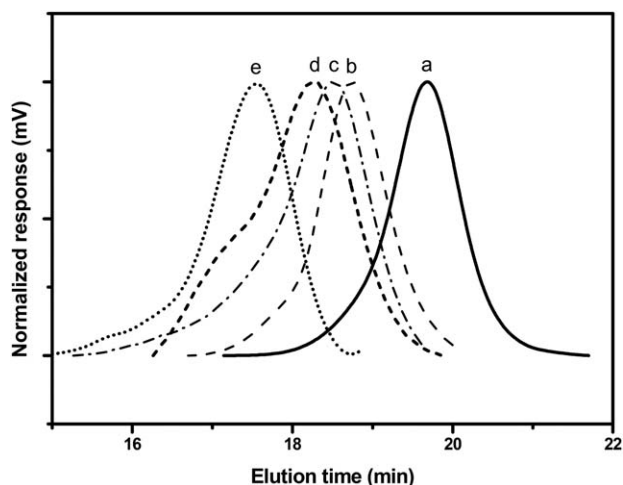


Figure 1. GPC plots for free PGMA during γ -initiated graft polymerization from PE/PP mediated via the RAFT agent CPPA at 10% (wt/wt) GMA and 1% (wt/wt) Tween20: (a) 0.5 kGy, 100:1 GMA-to-CPPA, $M_{n, GPC} = 2500$, PDI = 1.16; (b) 1 kGy, 100:1 GMA-to-CPPA, $M_{n, GPC} = 4900$, PDI = 1.16; (c) 3 kGy, 100:1 GMA-to-CPPA, $M_{n, GPC} = 6700$, PDI = 1.16; (d) 0.5 kGy, 400:1 GMA-to-CPPA, $M_{n, GPC} = 7800$, PDI = 1.29; (e) 1 kGy, 400:1 GMA-to-CPPA, $M_{n, GPC} = 13,600$, PDI = 1.28.

the monomer-to-RAFT agent ratio (i.e., low RAFT agent concentration, entries 4–5 and 9–10) or the irradiation time resulted in slight tailing which indicated the presence of high-molecular-weight polymers, which possibly came from the irreversible termination of the propagating chains. Nonetheless, the narrow molecular weight distribution, expressed as PDI, from the GPC trace of samples in Figure 1, is a clear indication that the main RAFT mechanism was established during grafting.

In contrast to the RAFT-mediated polymerizations, the conventional polymerizations were characterized by PGMA with very high M_n and broad molecular weight distribution as shown by entries 18–21 in Table I. GMA is a specialty monomer and it is difficult to polymerize via conventional radical polymerization without affecting its oxirane group. During conventional radical polymerization of GMA under γ -radiation, excessive crosslinking via the oxirane group and/or the radicals formed on the polymer chains leads to a gel material for which the application is restricted.^{22,26} However, there was no gel formation observed during the RAFT-mediated polymerization in this study. Rather than the high monomer conversion and high molecular weight polymers at the early portion of the polymerization, which characterize conventional grafting, the polymerization was slower and the resultant polymer had narrow PDI, especially at low targeted M_n values.

At higher targeted M_n , the PDI of the free PGMA were greater than 3, except at 10% monomer concentration (entries 4 and 5, 9 and 10). These data are consistent with the results gathered by other researchers.^{27,28} They suggested that these high PDI values were not due to colloidal instability but were most probably a result of heterogeneous distribution of RAFT agent among the micellar monomer droplets. When higher M_n values

were targeted, the lower amount of RAFT agent resulted in a greater number of micellar monomer droplets with different monomer-to-RAFT agent ratios, resulting to different polymerization rates and eventually leading to higher PDIs. At fixed monomer-to-RAFT agent ratio, the concentration of RAFT agent increased with increasing monomer concentration; hence, the PDI of free PGMA remained reasonably low at 10% monomer concentration even at 400:1 monomer-to-RAFT agent ratio.

The results listed in Table I show that M_n of PGMA increased with polymerization time and polymers with narrow molecular weight distribution were prepared under ^{60}Co γ -irradiation in the presence of CPPA. Figure 2 shows a typical ^1H -NMR spectrum of PGMA in CDCl_3 , with chemical shift assignments, obtained in the RAFT-mediated γ -radiation induced graft polymerization of GMA from PE/PP. The spectra collected in this study agree with those in the literature.^{29,30}

The observed ratio of the peak areas of (a), (b), and (d) against (e) is about 3:1; indicating that the oxirane groups in the PGMA remained intact during the γ -radiation induced graft polymerization in the presence of CPPA.³¹ The signals at $\delta = 7.3\text{--}8.0$ (δ , ppm: 7.90, d, 2H; 7.50, t, 1H; 7.35, t, 2H) correspond to the aromatic protons of dithiobenzoate aromatic hydrogens. Assuming that each macromolecule contains one molecule of dithiobenzoate, the $M_{n, \text{NMR}}$ of PGMA polymerized with CPPA can be estimated using the following equation:

$$M_{n, \text{NMR}} = \frac{I_{a,b}}{I_{7.3-8.0}} \times 142.15 + 279.38 \quad (3)$$

where $I_{a,b}$ and $I_{7.3-8.0}$ are the integral of the peaks at $\delta = 0.5\text{--}2.3$ (main chain repeating unit, 5H) and $\delta = 7.3\text{--}8.0$ (dithiobenzoate end-group, 5H), respectively, 142.15 and 279.38 are the molecular weight of GMA and CPPA, respectively. The NMR-based M_n are tabulated in Table I. The M_n 's derived from NMR agreed well with the theoretical molecular weights, $M_{n, \text{th}}$, indicating a controlled character of the polymerization in the presence of CPPA under γ -irradiation. Some entries for $M_{n, \text{GPC}}$ deviated largely from $M_{n, \text{th}}$ and $M_{n, \text{NMR}}$, but the observed trends were still similar. The observed discrepancies in M_n were probably due to the different hydrodynamic volumes of PGMA and polystyrene standards in THF solvent.

Differential scanning calorimeter (DSC) was used to study the thermal transitions in the course of heating under inert atmosphere. As shown in Figure 3, the glass transition temperature of free PGMA (PDI ≤ 1.38) homopolymers exhibited dependence toward M_n , wherein the T_g increased with increasing M_n . It is well known that the T_g of a given polymer is a function of its chain length.³² The PE/PP-g-PGMA with 92% Dg also exhibited a thermal transition at around 83 °C, higher temperature than its free PGMA counterpart ($T_g = 78$ °C). The increase in T_g of the grafted samples was attributed to the restricted mobility of PGMA chains grafted from PE/PP surface.⁴ This observation confirmed the covalent attachment of PGMA chains from the surface of PE/PP, and restriction imposed on the segmental motion of the PGMA chains as a result of grafting. PE/PP-g-PGMA with lower Dg's did not produce significant signals during the DSC analysis because of either dilution of the PGMA

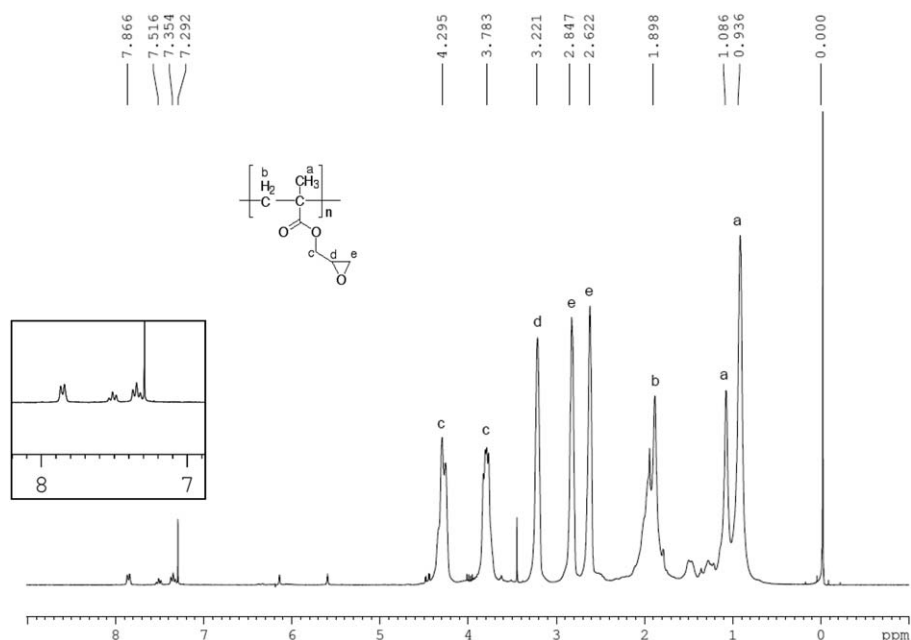


Figure 2. ^1H -NMR spectrum (300 MHz, CDCl_3) of free PGMA ($M_{n,\text{GPC}} = 4900$, $\text{PDI} = 1.16$) during γ -initiated graft polymerization from PE/PP mediated via the RAFT agent CPPA at 10% (wt/wt) GMA, 1% (wt/wt) Tween20 and 100:1 GMA-to-CPPA molar ratio. Inset is the plot for peaks between 7.3 and 8.0 chemical shifts due to dithiobenzoate aromatic protons of CPPA.

chains or insufficient amount of analyzed samples due to low bulk density of grafted PE/PP.

Characterization of the PE/PP-g-PGMA

The structure of pristine and PGMA grafted PE/PP samples were studied by FTIR spectroscopy. Figure 4 compares the ATR-FTIR spectra of pristine PE/PP and its PGMA copolymers with various Dg. The spectrum of pristine PE/PP trunk polymer has peaks characteristic of its olefinic nature: $-\text{CH}_2$ symmetric and asymmetric stretching vibrations at 2848 and 2915 cm^{-1} , respectively, and $-\text{CH}_2$ bending vibrations at 1472 cm^{-1} . After grafting with PGMA, new peaks that correspond to the

chemical groups present in the grafted polymer chains appear: $\text{C}=\text{O}$ stretching vibration at 1720 cm^{-1} , $\text{C}-\text{O}$ stretching vibration at 1120 cm^{-1} and the characteristic oxirane group vibrational modes at around 1250, 900, and 850 cm^{-1} . Similar characteristic peaks were also reported by other researchers.^{22,30} Besides the absorption frequencies of bonds from PGMA, peaks from PE/PP fibers were also observed from the FTIR spectrum of PE/PP-g-PGMA. The functional groups due to the RAFT agent present on the polymer chain ends were not observed because of their low concentration in the sample.

Further examination of the FTIR spectra of PE/PP-g-PGMA with different Dg values showed that the absorption bands due

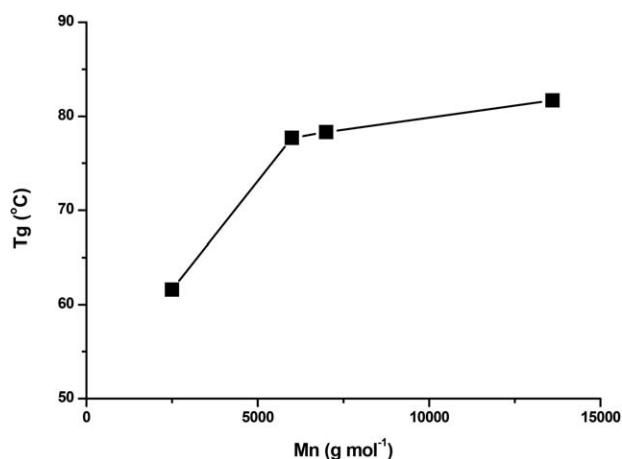


Figure 3. Dependence of T_g on molecular weight of free PGMA during γ -initiated graft polymerization from PE/PP mediated via the RAFT agent CPPA.

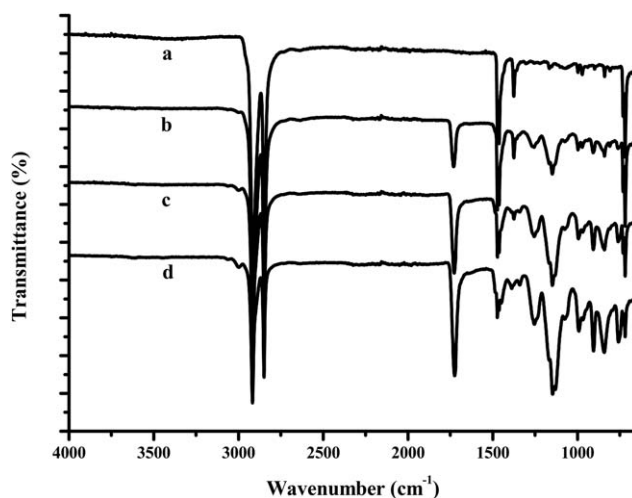


Figure 4. ATR-FTIR spectra of (a) pristine PE/PP, (b) PE/PP-g-PGMA, Dg = 7%, (c) PE/PP-g-PGMA, Dg = 24%, and (d) PE/PP-g-PGMA, Dg = 92%.

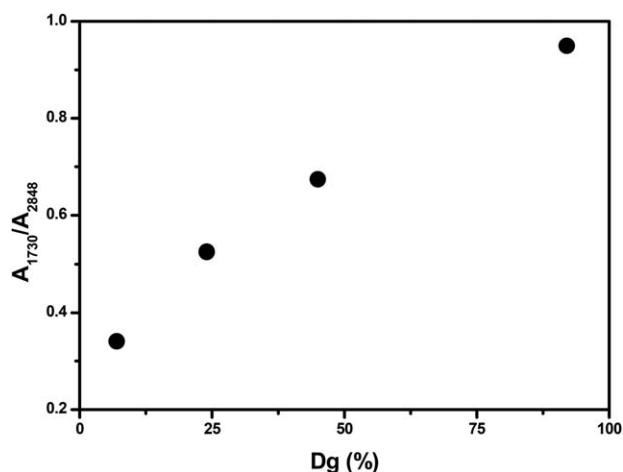


Figure 5. Plot of 1730 and 2848 cm^{-1} peak area ratios at different Dg. Peak areas were calculated from FTIR spectrum at absorbance mode.

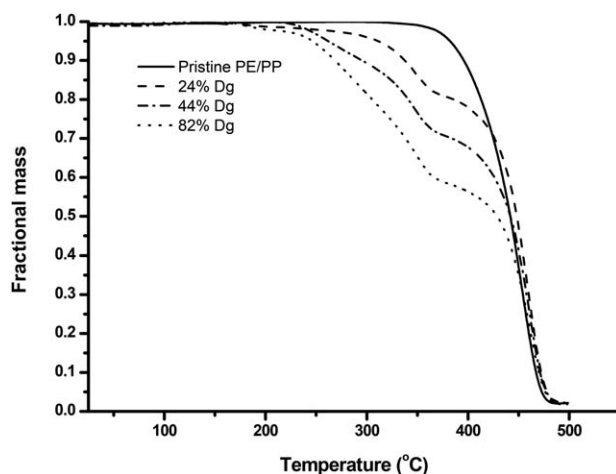


Figure 7. Thermogravimetric curves of pristine PE/PP and PE/PP-g-PGMA with different Dg.

to the C=O, C—O, and oxirane group increased with Dg, as shown in Figure 4(b–d). The observed trend was due to the increase in number of C=O, C—O, and oxirane chemical groups at the surface of the grafted material with increment in Dg and according to Beer–Lamberts law, the corresponding absorbance of these vibrational transitions must also increase. The peak at 2848 cm^{-1} , which is due to $-\text{CH}_2$ symmetric stretch remained almost unaffected by the increase in Dg. The plot of the ratio of peak areas (Figure 5), calculated in absorbance mode, at 1730 and 2848 cm^{-1} against different Dg of PE/PP-g-PGMA yielded a linear fit with $R^2 = 0.98$. This plot allows the evaluation of Dg of PE/PP-g-PGMA, which is prepared using the same RAFT-mediated graft polymerization technique described in this study, from its ATR–FTIR spectrum.

Figure 6(a–f) shows the SEM pictures and corresponding EDX oxygen elemental maps of pristine PE/PP and PE/PP-g-PGMA, with various Dg, prepared using RAFT-mediated graft polymerization. The pristine PE/PP was porous and composed of fibers with homogeneous morphology. The randomly oriented fibers were interconnected and after grafting, no homopolymer aggregation blocking the pores was observed. The oxygen map for the pristine PE/PP fibers shows the presence of oxygen in the material that was supposed to be comprised of carbon and hydrogen only. The detected oxygen atoms might have been incorporated on the fabric during the manufacturing process or due to surface oxidation. After γ -radiation induced RAFT-mediated grafting to 24% and 92% Dg, the fiber thickness noticeably increased as illustrated in Figure 6(b,c) in agreement

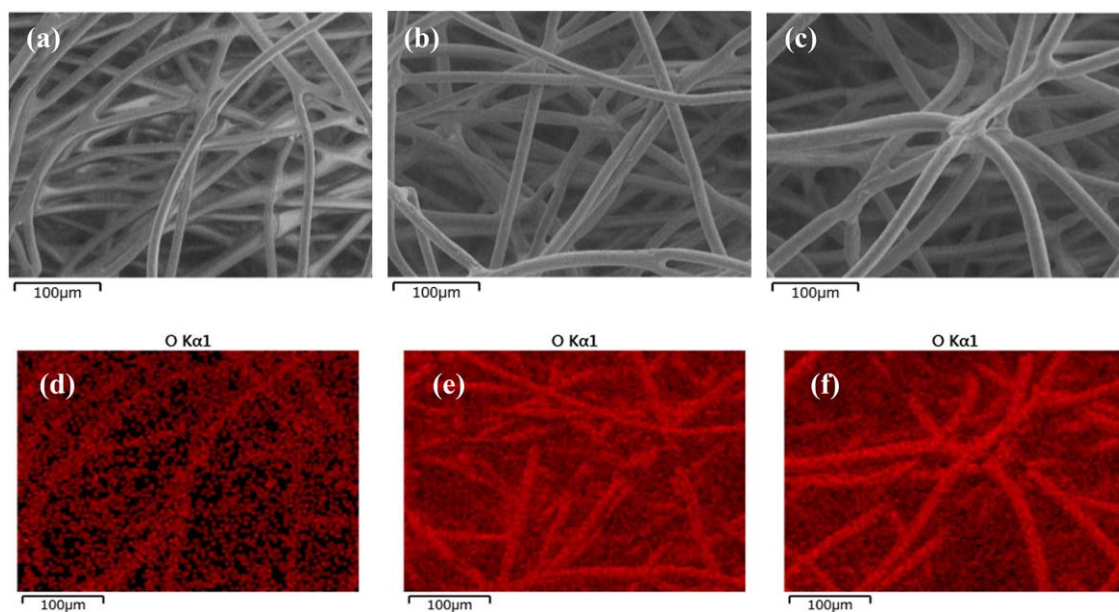


Figure 6. SEM images at 300 $\times$ magnification and elemental oxygen maps of (a,d) pristine PE/PP; (b,e) PE/PP-g-PGMA, Dg = 24%; and (c,f) PE/PP-g-PGMA, Dg = 92%. [Color figure can be viewed at wileyonlinelibrary.com]

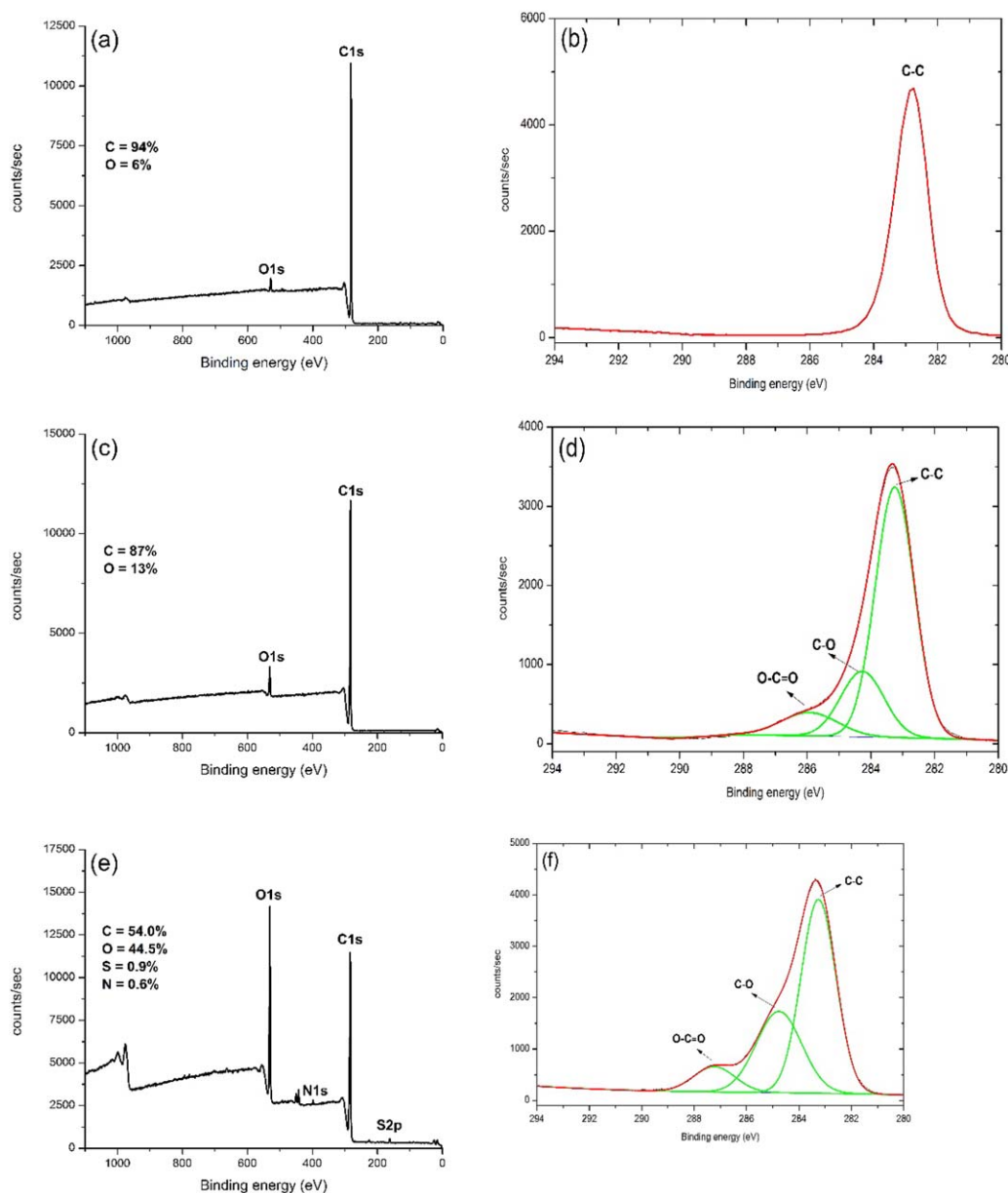


Figure 8. XPS survey wide scan and C1s high-resolution spectra of (a,b) pristine PE/PP, (c,d) PE/PP-g-PGMA, RAFT-mediated grafting $D_g = 90\%$, (e,f) PGMA homopolymer, RAFT-mediated, $M_{n,GPC} = 2500 \text{ g mol}^{-1}$, $PDI = 1.16$. [Color figure can be viewed at wileyonlinelibrary.com]

with other works.^{2,20,22} Moreover, PE/PP-g-PGMA exhibited higher oxygen density, as shown in Figure 6(e,f), compared to the pristine PE/PP. The increase in oxygen density was expected because the grafted PGMA chains contain oxygen atoms in its structure, i.e., from the oxirane and ester groups. These evidences showed that the grafting process was successfully carried out. The sulfur atoms from the RAFT agent ends on the grafted PGMA chains were not detected because of their very low concentration.

Introduction of PGMA chains on PE/PP trunk polymer via RAFT-mediated graft polymerization imparted new chemical bonds that modified the thermal behavior of PE/PP. Figure 7 demonstrates the thermal decomposition profile of pristine PE/

PP and PP-g-PGMA with various D_g values. The PE/PP base material showed a single decomposition profile with maximum weight loss at around 450°C , in close agreement with previous reports.^{20,33} In PE/PP-g-PGMA, a three-step decomposition profile was observed, with the last step coinciding with the decomposition profile of the PE/PP trunk polymer. The grafted material did not show significant mass loss before 180°C . The initial degradation step was attributed to the depolymerization from the unsaturated chain ends and the weight loss above 300°C was associated with degradation by random chain scission.³⁴ The fractional mass loss before 400°C increased with increasing D_g , a clear indication that the thermal decomposition and observed weight losses at lower temperatures were due to the PGMA graft chains.

Survey wide scan and C1s high resolution X-ray photoelectron spectra of pristine and grafted PE/PP and the PGMA homopolymer are shown in Figure 8. The surface elemental compositions, which were determined from peak areas of the XPS spectra, are given as insets in the survey scans. The wide survey scan spectra of pristine PE/PP indicated that it was composed of carbon and oxygen atoms. The oxygen atoms in the PE/PP trunk material, which is olefinic in nature, might have been produced from surface oxidation reactions which took place during processing and storage of the material. This observation corroborated with the SEM-EDX oxygen map data of pristine PE/PP presented in Figure 6(d). Nonetheless, the high resolution C1s spectrum of pristine PE/PP [Figure 8(b)] revealed that the C—C bonds were dominant in its structure, as expected from the olefinic nature of PE/PP. The oxygen content increased from 6% to 13% after grafting, as shown in Figure 8(c). Moreover, the deconvoluted high resolution C1s spectrum of PE/PP-g-PGMA [Figure 8(d)] showed the presence of other peaks besides the C—C peak. These peaks were attributed to the C—O and C=O bonds that were present in the PGMA graft chains. The presence of these new peaks and the PDI data from GPC confirmed the successful RAFT-mediated graft polymerization of emulsified GMA from PE/PP under gamma-ray irradiation.

Analysis of the PGMA homopolymer (i.e., free PGMA) signified that some of the chains were capped by either the sulfur- or nitrogen-containing portion of the CPPA RAFT agent, as indicated by the presence of S2p and N1s peaks in the XPS survey scan in Figure 8(e). The S2p peak was not observed from the wide scan spectrum of PE/PP-g-PGMA because of its low concentration on the surface. Furthermore, the atomic C/O ratio calculated after grafting, which was about 6.7, was higher than the theoretically calculated value from the PGMA structure where C/O should be equal to 2.3. This suggests that the PE/PP trunk polymer substantially contributed to the top 10 nm surface layer, i.e., the top surface layers were not solely constituted by PGMA graft chains. It is possible that some of the RAFT-mediated graft polymerization proceeded inside the surface layers by creating PGMA grafted zones that proceeded inward by the front mechanism.^{2,20,33}

CONCLUSIONS

PGMA, a versatile functional polymer, was grafted from PE/PP nonwoven fabric via RAFT-mediated graft polymerization under γ -irradiation. The application of γ -irradiation to emulsified GMA monomer in the synthesis of grafted polymers and its combination with RAFT-mediated polymerization combined powerful and industry-relevant techniques that allowed the facile synthesis of polymer surfaces modified with well-controlled grafted chains for various applications. Synthesis of grafted PE/PP with tunable Dg and almost uniform graft chains was achieved by varying the grafting parameters in the presence of RAFT agent. The reactive pendant oxirane group of GMA provides versatility to the PE/PP-g-PGMA which can be tailor-made to meet a wide variety of demanding specifications.

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